

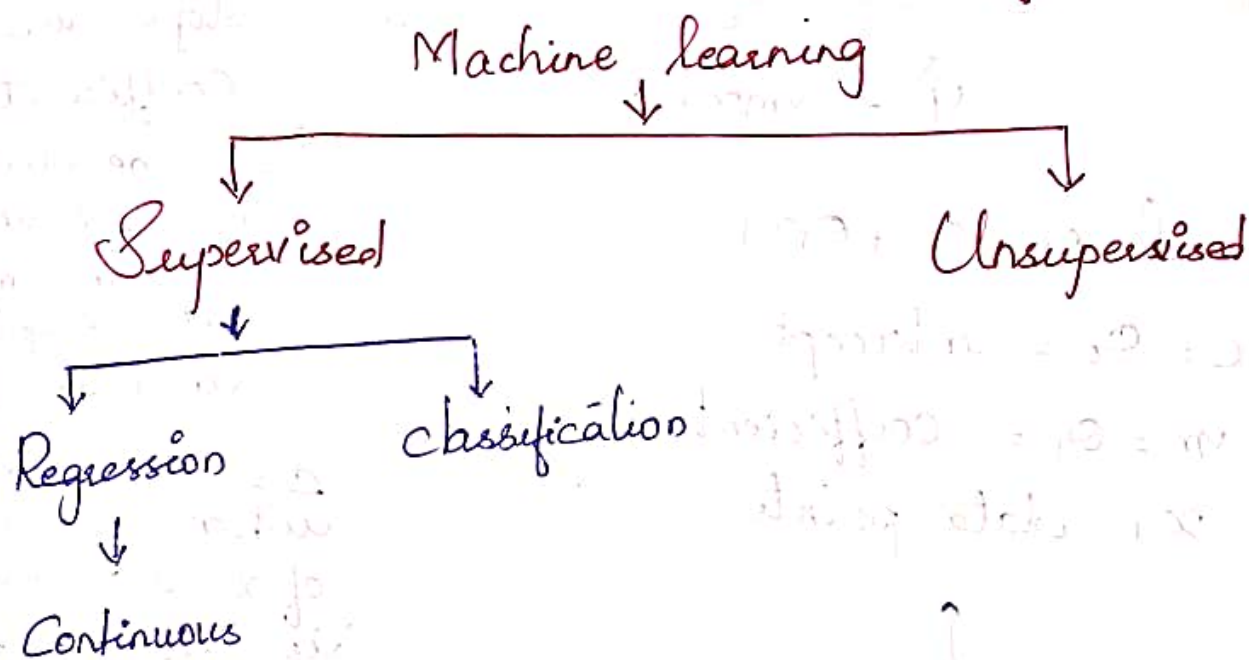
ML MACHINE LEARNING

UNIT. II

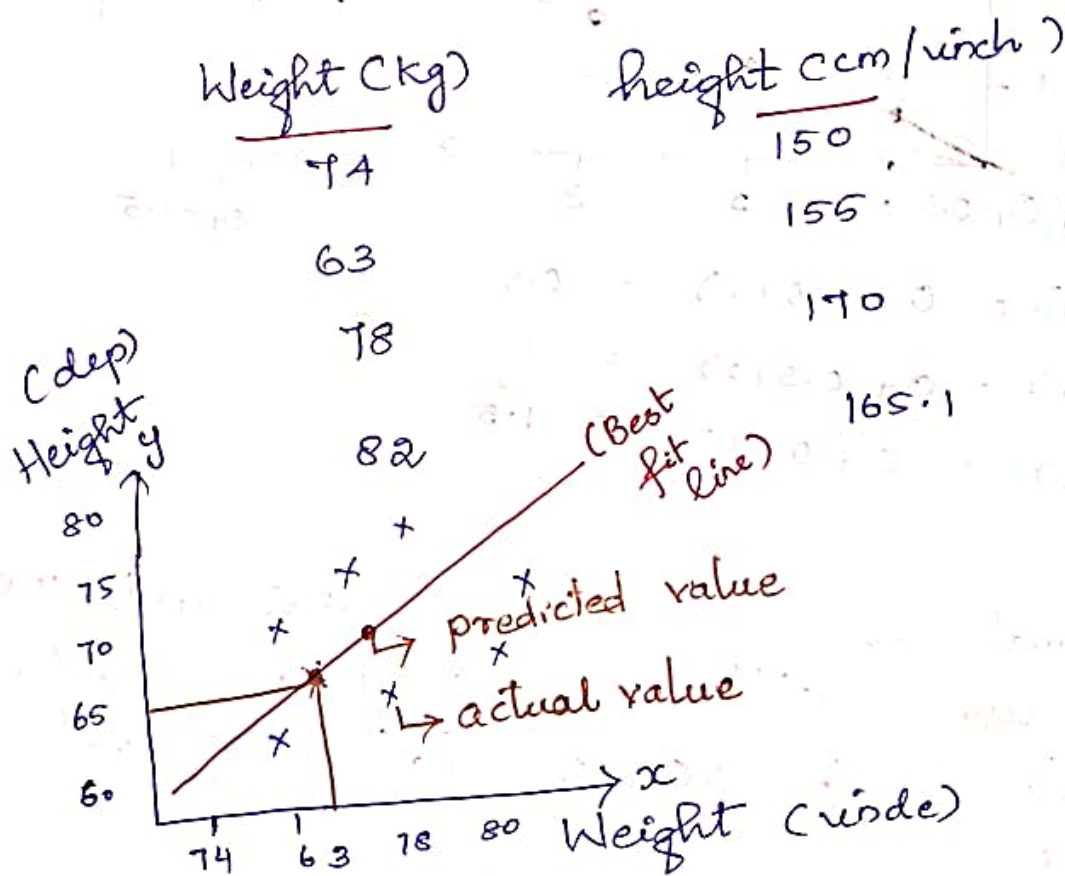
Supervised learning - Single & Multi variable cases, Regularization, Bias & Variance, Logistic Regression.

Classification - k. Nearest Neighbour - Naive Bayes, Decision tree, Random forest, Support Vector Machine, Case study on advanced supervised ML techniques for Regression & Classification.

SUPERVISED LINEAR REGRESSION.



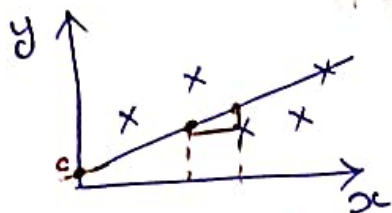
LINEAR REGRESSION



" BEST FIT LINE "

AIM: To find best fit line with
minimum error.

y is a linear function of x .



$$\hat{y} = mx + c$$

$\hat{y}$ = predicted o/p
 m = slope or
 Coefficient

"C same unit
 increase in x
 is how much
 it is going to
 increase in y"

$$h(x) = \theta_0 + \theta_1 x_1$$

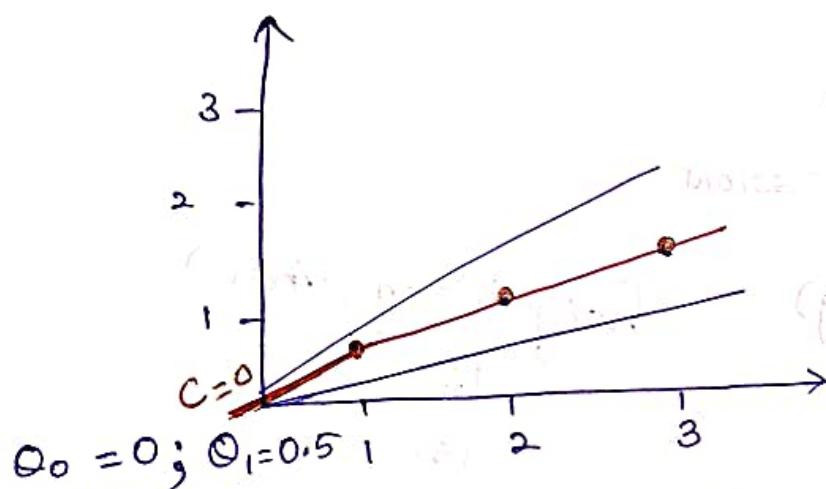
$C = \theta_0$ = intercept

$m = \theta_1$ = Coefficient

x_1 = data points.

C = intercept
 when your value
 of $x = 0$, where
 is the best fit
line intercepting y.

x = data point



$$h(1) = 0 + 0.5(1) = 0.5$$

$$h(2) = 0 + 0.5(2) = 1$$

$$h(3) = 0 + 0.5(3) = 1.5$$

$$\theta_1 = 1$$

$$\theta_1 = 1.5$$

"Ultimate aim is to find best fit line
 so we will be varying θ_0 & θ_1
 value is varied to get best
 fit line"

"we can use Cost Function"

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^i) - y^i)^2$$

cost function =

diff x^2
 $2x^2 = 2x$
 to find slope when θ value changes
 $\frac{1}{2} ()^2$
 $= \frac{x}{2}$
 to find slope it is easy.

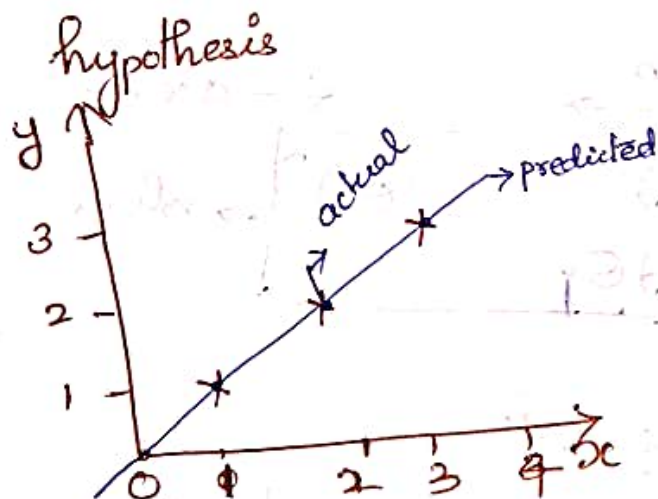
∴ There cost function or squared error fn is given by.

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^i) - y^i)^2$$

hypothesis

$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

Cost function



$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

$$\theta_0 = 0$$

$$\text{where } \theta_1 = 1$$

$$1 = 0 + 1 \times 1 = 1$$

$$2 = 0 + 1 \times 2 = 2$$

$$3 = 0 + 1 \times 3 = 3$$

$$J(\theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^i) - y^i)^2$$

$$= \frac{1}{6} [(0^2) + (0^2) + (0^2)]$$

$$= 0$$

$$J(\theta_1) = 0$$

$$\theta_1 = 1$$

$$J(\theta_1) = \frac{1}{2m} \sum_{i=1}^m (h(x^i) - y^i)^2 \quad \text{when } \theta = 1$$

$$= \frac{1}{6} (0^2 + 0^2 + 0^2)$$

$$= 0$$

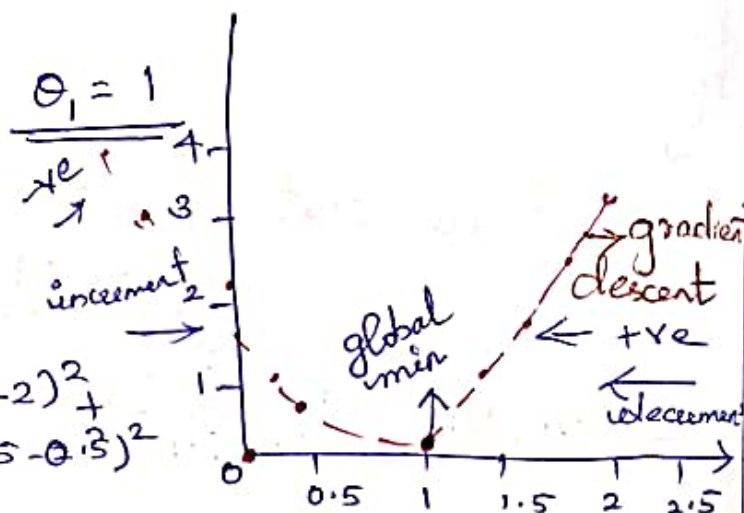
$$J(\theta_1) = 0 \quad \text{when } \theta_1 = 1$$

$$\text{when } \theta_1 = 0.5$$

$$J(0.1) = \frac{1}{6} [(0.5-1)^2 + (1-2)^2 + (1.5-0.3)^2]$$

$$= \frac{1}{6} (3.5)$$

$$J(\theta_1) = 0.58$$



Convergence theorem

$$\theta_j := \theta_j - \alpha \left[\frac{d}{d\theta_j} (J(\theta_1)) \right] \rightarrow \text{slope}$$

learning rate

$$\alpha = 0.01$$

→ slowly it will reach to global minimum.

Outline

1) start with θ_0 & θ_1

2) keep changing θ_0 & θ_1 to reduce $J(\theta_0, \theta_1)$ until to reach global

3) minima Convergence theorem

LINEAR REGRESSION

Simple linear Regression

$$\hat{y} = \theta_0 + \theta_1 x$$

$\hat{y}$ = predicted output (dependent)

θ_0 = Intercept / Bias

θ_1 = slope / Weight / coeff

x = Input (independent)

so find value of θ_0 & θ_1

Find Mean & Covariance.

$$\bar{x} = \frac{1}{N} \sum x_i \quad \bar{y} = \frac{1}{n} \sum y_i$$

$$\text{Variance}(x) = \frac{1}{n-1} \sum (x_i - \bar{x})^2$$

$$\text{Covariance}(x, y) = \frac{1}{n-1} \sum (x_i - \bar{x})(y_i - \bar{y})$$

$$\theta_1 = \frac{\text{Cov}(x, y)}{\text{Var}(x)}$$

$$\theta_0 = \bar{y} - \theta_1 \bar{x}$$

Problem : 01

x	y
1	1
2	2
3	1.3
4	3.75
5	2.25

$$n = 5 \quad \bar{x} = \frac{1}{5} (1+2+3+4+5)$$

$$= \frac{15}{5}$$

$$\boxed{\bar{x} = 3}$$

$$\bar{y} = \frac{1}{5} (1+2+1.3+3.75+2.25)$$

$$\boxed{\bar{y} = 2.06}$$

$$\text{Cov}(x, y) = \frac{1}{n-1} \sum (x_i - \bar{x})(y_i - \bar{y})$$

$$\boxed{\text{Cov}(x, y) = 1.0625}$$

$$\text{Var}(x) = \frac{1}{n-1} \sum (x_i - \bar{x})^2$$

$$\boxed{\text{Var}(x) = 2.5}$$

$$\theta_1 = \frac{\text{Cov}(x, y)}{\text{Var}(x)} = \frac{1.0625}{2.5}$$

$$\boxed{\theta_1 = 0.425}$$

$$\theta_0 = \bar{y} - \theta_1 \bar{x}$$

$$\theta_0 = 2.06 - (0.425)(3.0)$$

$$\boxed{\theta_0 = 0.785}$$

$$\therefore y = 0.785 + 0.425x$$

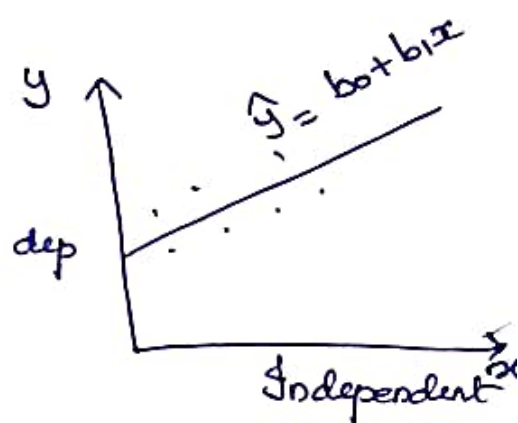
② LINEAR REGRESSION.

$$\hat{y} = b_0 + b_1 x$$

$\hat{y}$ = predicted output

b_0 = intercept

b_1 = slope.



when $x \uparrow$ $y \uparrow$ slope is (+ve)

$$\hat{y} = b_0 + b_1 x$$

when $x \downarrow$ $y \downarrow$ slope is (-ve)

Summation based formula, $\hat{y} = b_0 - b_1 x$

$$b_1 = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

Simple linear Regression - Relationship b/w 2 Variables
Multi linear Regression - More than 2 Variables.

$x \rightarrow$ independent variable (input)

$y \rightarrow$ dependent variable (output)

The regression line equation is

$$\hat{y} = b_0 + b_1 x$$

$\hat{y}$ = predicted value of y

b_0 = intercept

b_1 = slope / coefficient

slope

if $b_1 = 2$ y increases by 2 for every one unit increase in x

$b_1 = -3$ y decreases by 3 for every one unit increase in x

$x \uparrow \quad y \uparrow$ slope is +ve ($\hat{y} = b_0 + b_1 x$)

$x \uparrow \quad y \downarrow$ slope is -ve ($\hat{y} = b_0 - b_1 x$)

$$\hat{y} = b_0 + b_1 x$$

The y intercept can be written as

$$b_0 = \bar{y} - b_1 \bar{x}$$

The slope of regression can be rewritten as

$$b_1 = \frac{\sum xy - n \bar{x} \bar{y}}{\sum x^2 - n \bar{x}^2} ; b_1 = \frac{\overline{xy} - (\bar{x})(\bar{y})}{\overline{x^2} - (\bar{x})^2}$$

$$b_1 = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

Summation

$$\bar{x} = \frac{\sum x}{n} ; \bar{y} = \frac{\sum y}{n}$$

Regression Equation of Y on X

Y is ~~in~~dependent X is independent

$$y = b_0 + b_1 x ; \text{predict } y$$

if x = hours studied / hr

y = marks / 10

$$y = 40 + 5x$$

$$y = 40 + 5(6)$$

$$y = 70 \text{ marks}$$

if a student studies 6 hours

Regression equation of x on y

$$X = a_0 + a_1 Y$$

If we want to predict how many study hours are needed to score 70 marks.

X = dependent

Y = independent. ; predict x

Problem: 01

Consider an example where 5 weeks sales data (in thousands) is given below. Apply linear regression technique to predict the 7th & 12th week sales.

Week	Sales (thousands)
1	1.2
2	1.8
3	2.6
4	3.2
5	3.8

Linear Regression equation can be written as

$$Y = b_0 + b_1 X$$

$$b_1 = \frac{\overline{XY} - (\bar{X})(\bar{Y})}{\overline{X^2} - (\bar{X})^2}$$

X_i	Y_i	X_i^2	$X_i Y_i$
1	1.2	1	1.2
2	1.8	4	3.6
3	2.6	9	7.8
4	2.2	16	8.8
5	2.8	25	14
Σ	10.6	55	35.4

$$\bar{X} = \frac{\sum X_i}{n}$$

$$\bar{Y} = \frac{\sum Y_i}{n}$$

$$\overline{X^2} = \frac{\sum X_i^2}{n}$$

$$\overline{XY} = \frac{\sum X_i Y_i}{n}$$

$$\bar{X} = \frac{1+2+3+4+5}{5} = \frac{15}{5} = 3$$

$$\bar{Y} = \frac{1.2+1.8+2.6+2.2+2.8}{5} = 2.52$$

$$\overline{X^2} = \frac{1+4+9+16+25}{5} = 11$$

$$\overline{XY} = \frac{1.2+3.6+7.8+8.8+14}{5} = 8.28$$

$$\text{slope } b_1 = \frac{\bar{xy} - (\bar{x})(\bar{y})}{\bar{x}^2 - (\bar{x})^2}$$

$$= \frac{(8.88) - (3)(2.52)}{11 - 3^2}$$

$$b_1 = \frac{1.32}{2}$$

$$\boxed{b_1 = 0.66}$$

To compute intercept b_0 .

$$\boxed{b_0 = \bar{y} - b_1 \bar{x}}$$

$$b_0 = (2.52) - 0.66(3)$$

$$\boxed{b_0 = 0.54}$$

The regression line can be written as

$$\boxed{Y = 0.54 + 0.66x}$$

The predicted 7th week sale (when $x=7$)

$$Y = 0.54 + 0.66(7)$$

$$\boxed{Y = 5.16}$$

The predicted 12th week sale (when $x=12$)

$$Y = 0.54 + 0.66(12)$$

$$\boxed{Y = 8.46}$$

Problem : 02

Calculate the two regression equation of x on y & y on x from the data given below.

Price (Rs) 10 12 13 12 16 15

Amount demanded 40 38 43 45 37 43

Evaluate the likely demand when the price is 20.

Solution

x	y	x^2	y^2	xy
10	40	100	1600	400
12	38	144	1444	456
13	43	169	1849	559
12	45	144	2025	540
16	37	256	1369	592
15	43	225	1849	645

$$\bar{x} = 13$$

$$\bar{y} = 41$$

$$\bar{x}^2 = 173$$

$$\bar{y}^2 = 1689.3$$

$$\overline{xy} = 532$$

x = price independent

y = Amount (dependent)

Regression Eqn of x on y

$$x = b_0 + b_1 y$$

b_0 = intercept of x.

b_1 = slope.

$$b_1 = \frac{\overline{xy} - (\bar{x})(\bar{y})}{\bar{y}^2 - (\bar{y})^2}$$

$$b_1 = \frac{532 - (13)(41)}{1689.3 - (41)^2}$$

$$b_1 = -\frac{1}{8.3}$$

$$b_1 = -0.12048 \approx \boxed{b_1 = -0.121}$$

$$b_0 = \bar{x} - b_1 \bar{y}$$

$$b_0 = 13 - (-0.121 \times 41)$$

$$\boxed{b_0 = 17.961}$$

Regression Eqn of x on y

$$\boxed{x = 17.961 - 0.121y}$$

Regression Eqn of y on x

$$y = b_0 + b_1 x$$

b_0 = intercept of y

$$b_1 = \frac{\overline{xy} - \bar{x}\bar{y}}{\overline{x^2} - (\bar{x})^2}$$

$$= \frac{-1}{173 - (13)^2}$$

$$= -\frac{1}{4}$$

$$\boxed{b_1 = -0.25}$$

$$b_0 = \bar{y} - b_1\bar{x}$$

$$b_0 = 41 + 0.25(13)$$

$$\boxed{b_0 = 44.25}$$

Regression eqn of Y on X

$$Y = 44.25 - 0.25X$$

when x is 20 then y

$$Y = 44.25 - 0.25(20)$$

$$\boxed{Y = 39.25} //$$

Multiple

Take mean

$$y = b_0 + b_1x_1 + b_2x_2$$

$$\bar{y} = b_0 + b_1\bar{x}_1 + b_2\bar{x}_2$$

$$b_0 = \bar{y} - b_1\bar{x}_1 - b_2\bar{x}_2$$

Intercept $\therefore$ is chosen so that avg pred value equals avg actual value.

MULTIPLE Regression

Compute the multiple linear regression equation for the given data.

x_1	57	59	49	62	51	50	55	48
x_2	8	10	6	11	8	7	10	9
y	64	71	53	67	55	58	77	57

Soln :-

y	x_1	x_2	x_1^2	x_2^2	$x_1 y$	$x_2 y$	$x_1 x_2$
64	57	8	3249	64	3648	512	456
71	59	10	3481	100	4189	710	590
53	49	6	2401	36	2597	318	294
67	62	11	3844	121	4154	737	682
55	51	8	2601	64	2805	440	408
58	50	7	2500	49	2900	406	350
77	55	10	3025	100	4235	770	550
57	48	9	2304	81	2736	513	432
	431	69	23405	615	27264	4406	3762
62.75	53.81	8.65	2935.65	76.815	3408	530.75	470.25

To find Regression Coefficient.

$$b_1 = \frac{(\sum x_1 y)(\sum x_2^2) - (\sum x_1 x_2)(\sum x_2 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2}$$

$$b_1 = \frac{(27264)(615) - (3762)(4406)}{(23405)(615) - (3762)^2}$$

$$b_1 = 0.7952 \Rightarrow -0.1397$$

$$b_2 = \frac{(\sum X_2 Y)(\sum X_1^2) - (\sum X_1 X_2)(\sum X_1 Y)}{(\sum X_1^2)(\sum X_2^2) - (\sum X_1 X_2)^2}$$

$$b_2 = \frac{(4406)(23405) - (3762)(27264)}{24143}$$

$$b_2 = \frac{555262}{24143}$$

$$b_2 = 2.299$$

$$b_0 = \bar{Y} - b_1 \bar{X}_1 - b_2 \bar{X}_2$$

$$b_0 = (62.75) - (0.7952)(53.875) - (2.299)(8.625)$$

$$b_0 = 0.079125$$

The multiple linear reg eqn.

$$Y = 0.08 + 0.8X_1 + 2.3X_2$$

- 2) Compute the unknown value present in the given data using multiple regression

Age (x_1)	Year of exp (x_2)	Salary in (y) thous
35	1	20
25	1	10
40	2	35
50	4	?

X_1	X_2	Y	X_1^2	X_2^2	$X_1 Y$	$X_2 Y$	$X_1 X_2$
35	1	20	1225	1	700	20	35
25	1	10	625	1	250	10	25
40	2	35	1600	4	1400	70	80
$\frac{100}{3}$	$\frac{4}{3}$	$\frac{65}{3}$	3450	6	2350	100	140
33.33	1.33	21.67					

$$(\sum X_1^2) - \frac{(\sum X_1)^2}{n} = (\sum X_1 X_2) (\sum X_2 Y)$$

$$Q_1 = \frac{(\sum x_1 y)(\sum x_2^2) - (\sum x_1 x_2)(\sum x_2 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2}$$

$$b_1 = \frac{(2350)(6) - (140)(100)}{(3450)(6) - (140)^2}$$

$$b_1 = \frac{100}{1100} \quad \boxed{b_1 = 0.09}$$

$$b_2 = \frac{(\sum X_2 Y)(\sum X_1^2) - (\sum X_1 X_2)(\sum X_1 Y)}{(\sum X_1^2)(\sum X_2^2) - (\sum X_1 X_2)^2}$$

$$b_2 = \frac{(100)(3450) - (140)(2350)}{1100}$$

$$b_2 = \frac{16000}{1100} = 14.5$$

$$\boxed{b_2 = 14.5}$$

$$\bar{y} = b_0 + b_1 \bar{x}_1 + b_2 \bar{x}_2$$

$$b_0 = \bar{y} - b_1 \bar{x}_1 - b_2 \bar{x}_2$$

$$b_0 = (21.67) - (0.09)(33.33) - (14.5)(1.33)$$

$$b_0 = 21.67 - 2.999 - 19.285$$

$$\boxed{b_0 = -0.614}$$

$$Y = b_0 + b_1 x_1 + b_2 x_2$$

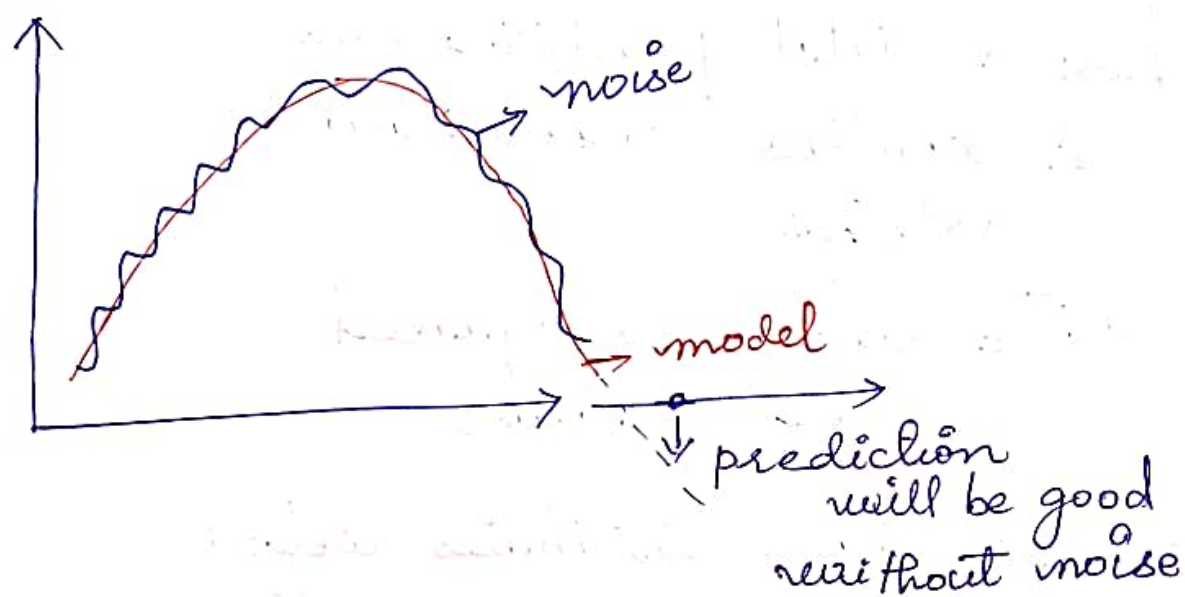
$$= -0.614 + 0.09 x_1 + 14.5 x_2$$

$$Y = -0.614 + 0.09(50) + 14.5(4)$$

$$\boxed{Y = 61.886}$$

Regularization

- used in ML to prevent overfitting
- improves model's ability to generalize to new & unseen data
- add the penalty to loss function
(overcomplex/noise training discourage)
- It simplifies the model by constraining its parameters, encouraging it to focus on the most imp patterns in data.



- Regularize loss = Original loss + λ penalty.
if $\lambda = 0$ you will get original loss
if $\lambda = \text{large}$ penalty or loss is more.

RIDGE REGULARIZATION. (L2)

→ modifies the linear regression loss function by adding an L2 penalty (sum of sq weights)

$$L_{\text{ridge}} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 + \lambda \sum w_i^2 \quad \boxed{m^2}$$

$$\boxed{L_{\text{ridge loss}} = \text{MSE} + \lambda \sum w_i^2} \quad \begin{matrix} \text{[} w_i \text{ slope] } \\ \hat{y} = mx + c \end{matrix}$$

$\downarrow$ slope²

Loss = total predicted error
 λ controls regularization strength

w_i^2 = represents squared weight values

→ The L2 penalty distributes weight values more evenly & stabilizes training.

if $\lambda = 0$ reduces to normal linear regression

λ large the model shrinks all weights close to 0 to prevent overfitting → not exactly 0

$$\hat{y} = \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + C$$

$$\hat{\theta} = \frac{m_1^2 + m_2^2 + m_3^2}{\theta_1^2 + \theta_2^2 + \theta_3^2}$$

$$\Delta = \lambda m^2$$

LASSO REGULARIZATION (L1)

$$L_{\text{lasso}} = \text{MSE} + \lambda \sum |w_i|^{|\text{slope}|} \quad /m)$$

$\lambda = 0$ behaves like linear reg

$\lambda = \text{large}$ model shrinks some weights to be exactly zero.

Effectively performing feature selection

$$\hat{y} = \cancel{\theta_1 x_1} + \theta_2 x_2 + \theta_3 x_3 + C$$

if $\theta_1 = 0$ you are reducing features

why L_2 is close to 0 ?
 L_1 is exactly to 0 ?

LOGISTIC REGRESSION.

to understand logistic regression

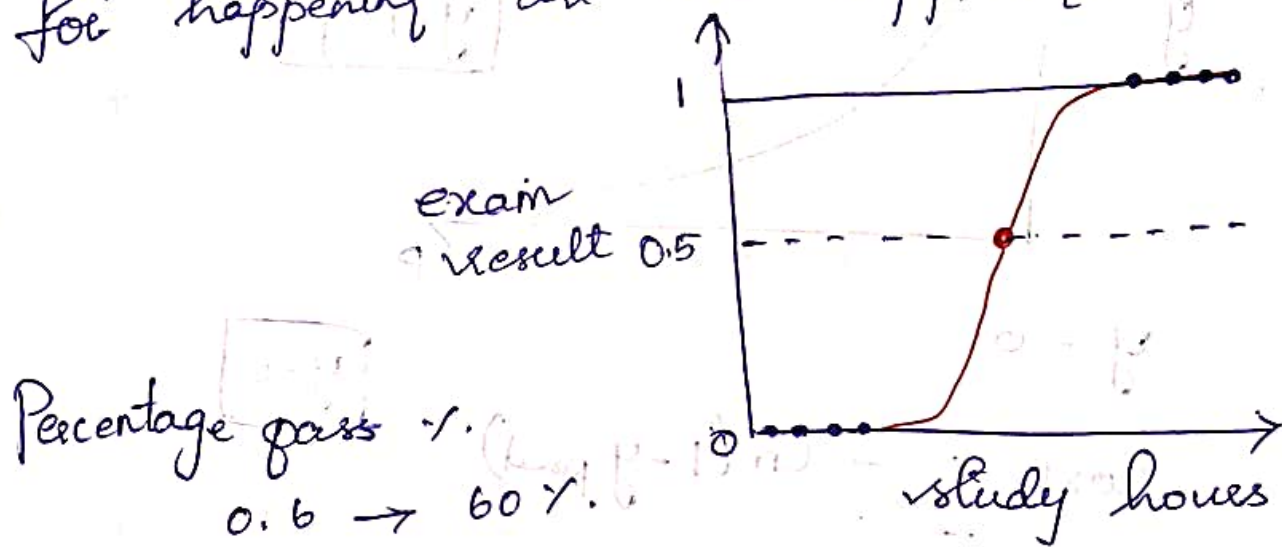
Prob of event happening.

$$\text{Odds}(p) = \frac{\text{prob of event happening}}{\text{prob of event not happ}}$$

if a person studies
for a hours weather
he will pass or fail $1-p$

↓ The value of odd ranges from

so probability plays very important role
for happening or not happening.



Evaluating model.

close to 1 → when outcome is 1

close to 0 → when outcome is 0.

Cost function can be used to find
weather our model is working best.

$$y=1$$

$$\text{Cost} = (-y_{\text{actual}}) \log(y_{\text{pred}}) -$$

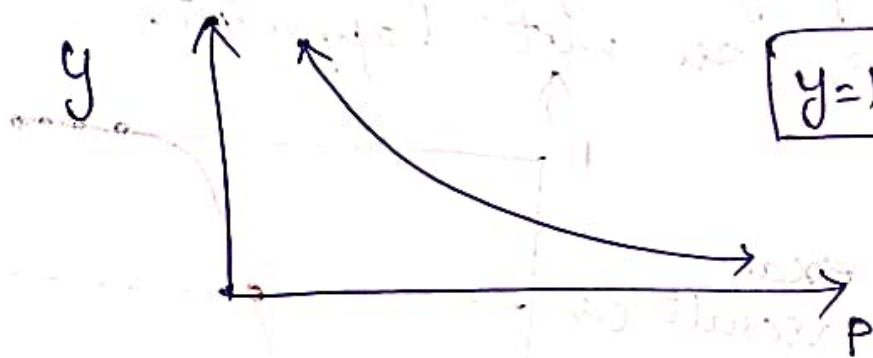
$$(1 - y_{\text{act}}) \ln(1 - y_{\text{pred}})$$

$$y=0$$

$$\text{Cost} = -\ln y_{\text{pred}}^{(p)} \quad y_{\text{act}} = 1$$

$$= \frac{-\ln(1 - y_{\text{pred}})}{(p_{\text{rob}})} \quad y_{\text{act}} = 0$$

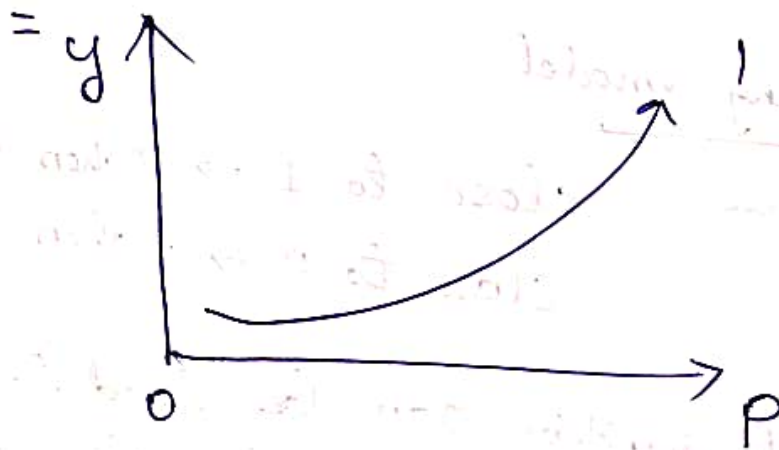
$$= -\ln(p)$$



$$y=0$$

$$\text{Cost} = -\ln(1 - y_{\text{pred}})$$

$$y=0$$



y_a y_p

MSE

$$(1 - 0.95)^2 = 0.0025$$

$$(1 - 0.1)^2 = 0.81$$

$$\therefore \text{cost} = -\ln(y_p)$$

$$= -\ln(0.95)$$

$$= 0.051$$

$$= -\ln(0.1)$$

$$= 2.302$$

→ penalizes more.

$$\text{log-loss} = \sum_{i=0}^n - (y_i * \log(p_i) + (1 - y_i) * \log(1 - p_i))$$

↑
↑
 class 1 class 0

$$\boxed{\hat{z} = wx + c}$$

$w =$ [weight

$x \rightarrow$ input

$c \rightarrow$ intercept.

PROBLEM

The data of pass or fail in an exam of 5 students are given below. Use logistic regression as classifier to answer the following

hours study	(1) pass or fail (0)
29	0
15	0
33	1
28	1
39	1

- i) Calculate the probability of pass for students who studied 33 hours.
- ii) At least how many hrs students should study that makes he will pass the course with probability more than 95%. Assume the model suggested by the optimizer will become the following odds of passing a score. for odds of passing the course is.

$$\log(\text{odds}) = -mx + c$$

$$-c + mx$$

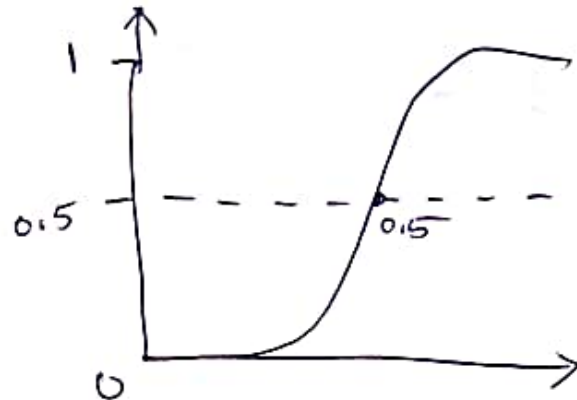
$$x = 33$$

$$\log(\text{odds}) = -64 + 2 \text{ hrs.}$$

Solution

$$S(x) = \frac{1}{1+e^{-x}}$$

$$P = \frac{1}{1+e^{-x}}$$



$$P = \frac{1}{1+e^{-2}}$$

② $P = 0.88$

$$z = -64 + 2 \times 33$$

① $z = -64 + 66$

$z = 2$

If students studies 33 hrs 88% chance to pass.

$$P = \frac{1}{1+e^{-z}} = 0.95$$

$$0.95(1+e^{-z}) = 1$$

$$0.95 + 0.95e^{-z} = 1$$

$$0.95e^{-z} = 1 - 0.95$$

$$e^{-z} = \frac{0.05}{0.95} = 0.0526$$

$e^{-z} = 0.0526$

$$\ln(e^x) = x$$

$$\ln(e^{-z}) = \ln(0.0526)$$

$$-z = -2.94$$

$z = 2.94$

$$2.94 = -64 + 2 \text{ hrs.}$$

$$2.94 + 64 = 2 \text{ hrs.}$$

$$\frac{66.94}{2} = \text{hours}$$

$$\boxed{\text{hours} = 33.47 \text{ hrs}}$$

Should
atleast ^{study} 33.47 hrs, so that he will
pass by 95%.
—————x—————

$$y = mx + c.$$

logistic prob.

To understand the logistic: let's start
odds of success

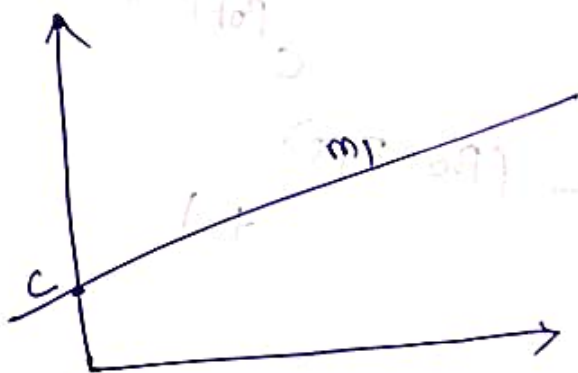
$$\text{odds}(P) = \frac{\text{Prob of an event happening}}{\text{prob. of an event not happ}}$$

$$\text{odds}(p) = \frac{p}{1-p} = \frac{0}{1} = 0; \frac{1}{1-1} = \frac{1}{0} = \infty$$

The value of probability ranges from

$$0 \rightarrow 1$$

and odd value ranges from $0 \rightarrow \infty$.



$$y = m_1x + c$$

or

$$y = \beta_0 + \beta_1x$$

$$\log \left(\frac{p(x)}{1-p(x)} \right) = \beta_0 + \beta_1x$$

$$\frac{p(x)}{1-p(x)} = e^{\beta_0 + \beta_1x}$$

$$p(x) = [1 - p(x)] e^{\beta_0 + \beta_1x}$$

$$p(x) = \frac{e^{\beta_0 + \beta_1x}}{1 + e^{\beta_0 + \beta_1x}}$$

$$P(x) + e^{\beta_0 + \beta_1 x} = e^{\beta_0 + \beta_1 x}$$

$$P(x) = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}$$

÷ Numerator & denom by $e^{\beta_0 + \beta_1 x}$

$$= \frac{1}{e^{\beta_0 + \beta_1 x} + e^{\beta_0 + \beta_1 x}}$$

$$= \frac{1}{e^{\beta_0 + \beta_1 x} + e^{\beta_0 + \beta_1 x}}$$

$$= \frac{1}{e^{-(\beta_0 + \beta_1 x)} + 1}$$

$$\therefore P(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

P.T.O

[Back]

K-Nearest Neighbour

For below given data set analyse given data set. a query is given as

Maths = 6, CS = 8 weather they will fall in pass or fail category.

Step 1

Assume $k=3$.

	Maths	CS	Result
1	4	3	Fail
2	6	7	Pass
3	7	8	Pass
4	5	5	fail
5	8	8	pass.

Step 2

Euclidean dist formula.

x_2, y_2
→ observed values

x_1, y_1
→ actual values.

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

1. $\sqrt{(6-4)^2 + (8-3)^2} = \sqrt{29} = 5.38$

2. $\sqrt{(6-6)^2 + (8-7)^2} = 1 \rightarrow$ nearest dist $\rightarrow$ pass

3. $\sqrt{(6-7)^2 + (8-8)^2} = 1 \rightarrow$ pass

4. $\sqrt{(6-5)^2 + (8-5)^2} = \sqrt{10} = 3.16$

5. $\sqrt{(6-8)^2 + (8-8)^2} = 2 \rightarrow$ pass

As per result 2, pass, 3, 5 $\rightarrow$ pass.

3 pass & 0 fail prob of pass is more.

2) Using K-NN predict the class of new data only with brightness = 20 saturation = 35.

Brightness	Saturation	Class	Euclidean dist.
40	20	Red	25
50	50	Blue	33.5
60	90	Blue	68
10	25	Red	14.14
70	70	Blue	61.03
60	10	Red	41.18
25	80	Blue	45.27
20	35	? RED.	Majority K=3.

step 1

$$K \text{ factor} = \sqrt{n}$$

$$= \sqrt{7}$$

$$K = 2.645$$

$$\boxed{K=3}$$

step 2 Observed = (20, 35)

$$EDI = \sqrt{(40-20)^2 + (20-35)^2}$$

$$\boxed{EDI = 25}$$

step 3. Ranking $k=3$.

3 Nearest point.

$k \in (14.14) \rightarrow \text{Red}$

25 $\rightarrow \text{Red}$

33.5 $\rightarrow \text{Blue}$.

Majority Voting

Red.

//

BAYE'S THEOREM

LOGISTIC REGRESSION. (50%)

A university wants to predict whether a student will pass (1) or fail (0) an exam based on the number of hours studied. where $\beta_0 = -5$ $\beta_1 = 1.5$

Student

hours studied

$m =$ Passed

A	1	0
B	2	0
C	3	0
D	4	1
E	5	1
F	6	1

Identify dependent & independent variable.

dependent $\rightarrow$ pass or fail

independent $\rightarrow$ hours studied

- ii) Write logistic regression model using sigmoid fn. (Assume fitted logistic reg model)

$$P(Y=1|x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

$\downarrow$ Outcome $\rightarrow$ indep

This model is used because the Outcome is binary pass/fail

- i) Calculate the prob of passing for a student studying for 3 hours & 4 hours.

3 hours

$$Z = \beta_0 + \beta_1 x$$

$$= -5 + 1.5(3)$$

$$= -5 + 4.5$$

$$Z = -5 + 1.5(3)$$

$$Z = -5 + 4.5$$

$$\boxed{Z = -0.5}$$

$$P(Y=1|x) = \frac{1}{1 + e^{-Z}}$$

$$e^{0.5} = 1.65$$

$$= \frac{1}{1 + e^{0.5}} = \frac{1}{1 + 1.65} = 0.38$$

fail

7 hours

$$Z = -5 + 1.5(4) = 1$$

$$P = \frac{1}{1 + e^{-1}} = \frac{1}{1 + 0.37}$$

$P = 0.73$ Pass

BAYE'S PROB.

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

NAIVE Bayes.

Make a naive assumption that features such as male, class, age, cabin, fare etc are independent of each other. or may not be.

Spam classification → Buy, cheap, prize

all features are created independent.

NAIVE BAYES.

	Spam		No spam.	
Total	25		75	
Buy	20	$\frac{20}{25} = \frac{4}{5} = 0.8$	5	$\frac{1}{15}$
Cheap	15	$\frac{3}{5}$	10	$\frac{2}{15}$
Buy & cheap (indep)	12	$\frac{12}{25} = \frac{12}{25}$	2/3	$\frac{2}{225}$

Problem 1: PPT

Pbm: 2.

Using Naive Bayes classifier find out the species class for given new instance

(Green, Leg=2, height=short & smelly=N)

S.No	① Colour	② Leg	③ Height	④ Smelly	Species
1.	white	3	short	Yes	M
2.	Green	2	Tall	No	M
3.	Green	3	short	Yes	M
4.	white	3	short	Yes	M
5.	Green	2	short	No	H
6.	White	2	Tall	No	H
7.	white	2	Tall	No	H
8.	White	2	short	Yes	H

Step 1: Convert the data into freq table

①

Colour	M	H	Prob
white	2	3	$P(W) = \frac{5}{8}$
Green	2	1	$P(G) = \frac{3}{8}$

Total = 8

②	Leg	M	H	Prob
	2	1	4	$P(\text{leg}=2) = \frac{5}{8}$
	3	3	0	$P(\text{leg}=3) = \frac{3}{8}$

Total = 8

③	heights	M	H	prob
	Tall	1	2	$P(T) = \frac{3}{8}$
	short	3	2	$P(S) = \frac{5}{8}$

④	Smelly.	M	H	$P(\text{Yes}_{\text{Sme}}) = \frac{4}{8}$
	Yes	3	1	
	No	1	3	$P(\text{No}_{\text{Sme}}) = \frac{4}{8}$

Step 2 Generate likelihood table by finding probs of each features

Color	M (†)	H (†)
white	$\frac{2}{4}$	$\frac{3}{4}$
green	$\frac{2}{4}$	$\frac{1}{4}$

legs	M (†)	H (†)
2	$\frac{1}{4}$	1
3	$\frac{3}{4}$	$\frac{0}{4}$

Height	M	H
Tall	$\frac{3}{4}$	$\frac{2}{4}$
short	$\frac{3}{4}$	$\frac{2}{4}$

Smelly.	M	H
Yes	$\frac{3}{4}$	$\frac{1}{4}$
No	$\frac{1}{4}$	$\frac{3}{4}$

step 2 To find the posterior prob based on bayes theorem.

New inst Color = green; Leg = 2; Height = Tall; Smelly = No

$$P(M | \text{new inst}) = \frac{P(M) * P(\text{Green} | M) * P(\text{leg} = 2 | M) * P(\text{Height} = \text{Tall} | M) * P(\text{Smelly} = \text{No} | M)}{P(\text{New inst})}$$

"Since denominator is same for all classes it can be ignored"

$$= 0.5 * \left(\frac{2}{4}\right) * \left(\frac{1}{4}\right) * \left(\frac{3}{4}\right) * \left(\frac{1}{4}\right)$$

$\frac{10}{5}$ vs $\frac{6}{5}$
which is better

$$P(M | \text{New inst}) = 0.011718$$

$$P(H | \text{New inst}) = \frac{P(H) * P(\text{Green} | H) * P(\text{leg} | H) * P(\text{Height} = \text{Tall} | H) * P(\text{Smelly} | H)}{P(\text{New inst})}$$

$$= 0.5 * \left(\frac{1}{4}\right) * (1) * \left(\frac{2}{4}\right) * \left(\frac{3}{4}\right)$$

$$P(H | \text{New inst}) = 0.046875$$

$$0.046875 > 0.011718$$

$$\text{Final output} = H$$